

Formulae

Please note that the present implementation differs to some extent from the original applications of the metrics because the parameters were (manually) set by the original authors to optimise the output of a given metric. This would not provide a reliable basis for comparison. In fact, setting the parameters to improve the accuracy of a metric gives information about the level of accuracy that can be reached with that metric on a given test set, but it gives little objective information about the ability to evaluate spectral stability by the metric *per se*.

1. Lennig (1978)

It computes an instability score, that is, the coefficient of change, where c_f is the coefficient of change of frame f .

$$c_f = \frac{|F1_f + F1_{f-1}| + |F1_f + F1_{f+1}|}{F1_f} + \frac{|F2_f + F2_{f-1}| + |F2_f + F2_{f+1}|}{F2_f} \quad (0.1)$$

2. Hillenbrand et al. (1995)

It consists of selecting the frames with the lowest slope in the $\log F2 - \log F1$ space, where slope_f is the slope in $\log F2 - \log F1$ space of frame f , and where sf stands for the start of frame f and ef the end of frame f .

$$\text{slope}_f = \frac{\log F2_{ef} - \log F2_{sf}}{\log F1_{ef} - \log F1_{sf}} \quad (0.2)$$

3. Van Bergem (1988)

It computes an instability score, where pv_f is the pooled within-variance of the log-transformed formant values within frame f .

$$pv_f = \frac{(n-1) \cdot \sum_{i=1}^n \frac{(\log F1_i - \overline{\log F1})^2}{n} + (m-1) \cdot \sum_{i=1}^m \frac{(\log F2_i - \overline{\log F2})^2}{m}}{n + m + p - 3} + \frac{(p-1) \cdot \sum_{i=1}^p \frac{(\log F3_i - \overline{\log F3})^2}{p}}{n + m + p - 3} \quad (0.3)$$

4. Van Bergem (1993)

It consists of the within standard deviation for the frame interval

whose center is frame f), where N stands for the number of cepstral coefficients calculated within a frame interval, c_{ij} represents the j th cepstral coefficient of frame i , and $\bar{c}_j$ is the mean value of all the j th cepstral coefficients of the frame interval.

$$V(f) = \sqrt{\frac{1}{N} \sum_{i=f-5}^{f+5} \sum_{j=1}^8 (c_{ij} - \bar{c}_j)^2} \quad (0.4)$$

Articulatory Targets Specification

The articulatory target specifications are adapted from Boersma (1998) and the PRAAT scripts available on its companion website.

To synthesize the CVC sequences:

- A contraction of the lungs is synthesized by setting the Lung parameter to 0.2 at 0 ms and to 0.0 after 100 ms to generate sufficient lung pressure, triggering an increase in oral pressure necessary for phonation and stop production (see Boersma, 1998, p. 128). This parameter is consistent for all stimuli.
- To generate an [a]-like sound, vocal fold vibration is produced by setting Interarytenoid to 0.5. Avoiding nasalization requires setting LevatorPalatini to 1.0 to close the nasal port.
- Similarly detailed steps for [b], [p], [m], [d], [t], [n], [g], [k], and [ŋ]-like sounds with corresponding parameter adjustments.

Timing Targets Specification

The sequences are built with specific points:

- **OccC1**: Point where all articulators for C1 have reached their target.
- **TsOccC1**: Time just before OccC1.
- **TeOccC1**: Time after OccC1 when the articulator activity for C1 starts transitioning towards the vowel target.

- **TsV**: Vowel target reached, maintained up to TeV.
- **TeV**: End of the vowel stable portion.

Timings:

- Start: always at 0 s;
- Onset: between Start and OccC1;
- TsOccC1: between Onset and OccC1;
- OccC1: between Start and TsV;
- TeOccC1: between OccC1 and TsV;
- TsV: after 0.05 s and before 0.25 s;
- TeV: after TsV + 0.05 s and before 0.3 s;
- TsOccC2: between TeV and OccC2;
- OccC2: between TeV and End;
- TeOccC2: between OccC2 and Offset;
- Offset: between OccC2 and End;
- End: always 0.35 s.